TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Chemistry Laboratory

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Full Solution of Exercise 9

	LSDA/6-31G(d)	LSDA/6-31G	Experimental Values
O-H distance, Å	0.983	0.995	0.965
O-O distance, Å	1.436	1.518	1.464
<(O-O-H), degrees	99.9	97.7	99.6
<(H-O-O-H), deg.	117.1	ca. 180	111.8

a. Polarization functions have a strong effect on molecular geometry, especially on the O-O distance and the dihedral angle.

b. The dihedral angle is completely wrong without polarization functions.

   Rotations about a single bond usually have potential curves with very shallow minima (unless there is sterical hindrance), and there may be multiple minima which differ only little in energy. Thus equilibrium dihedral angles are often hard to find. Atomic orbitals are distorted (polarised) upon molecule formation. This polarisation is allowed for by the use of polarisation functions. The correct angular shape of the atom in the molecule can be expressed much better using such functions, hence the shape of the rotational potential curve and the equilibrium dihedral angle can be better described.



[General help]



[To overview]



[Compute]



[Contact the course supervisor]

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